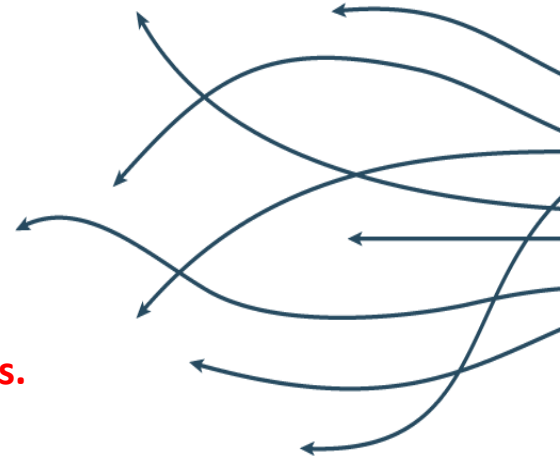


Visual Object Detection - I

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**Most slides are adapted from related courses or tutorials.
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北京大学
PEKING UNIVERSITY

AI and Zorro's Mask



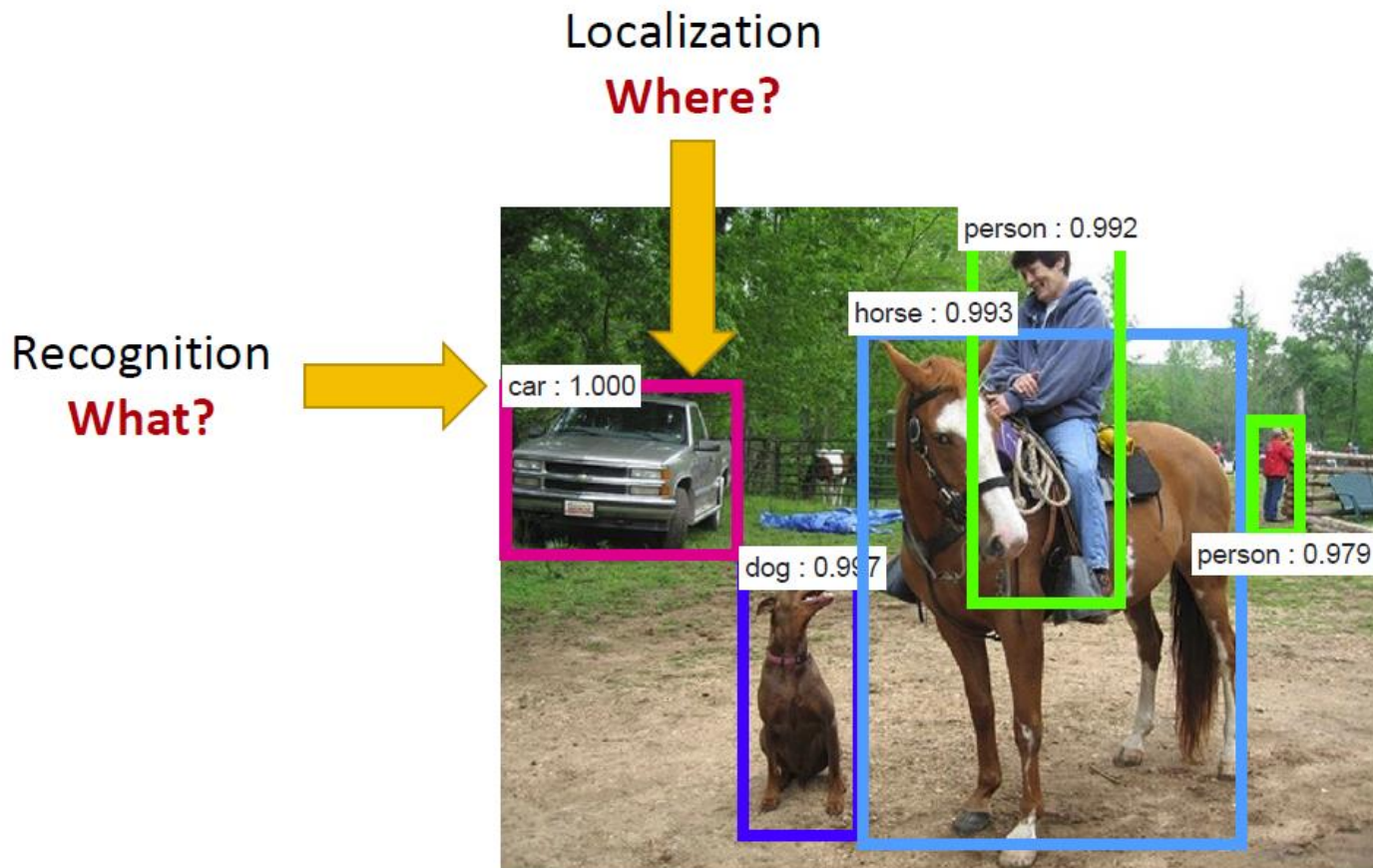
VS



Outline

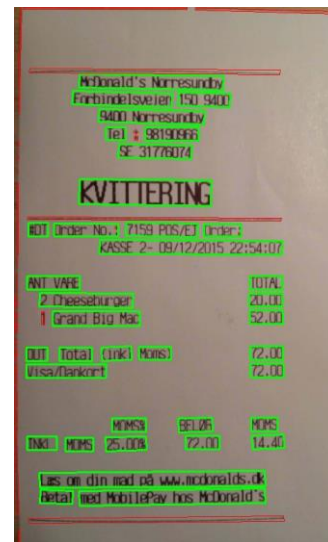
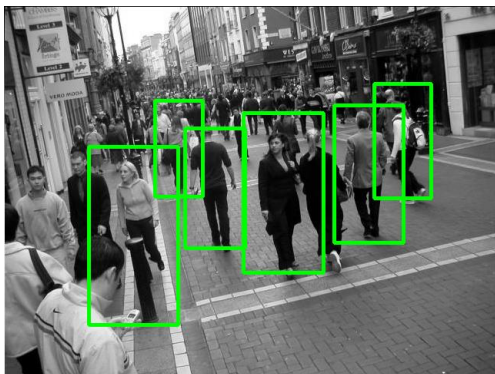
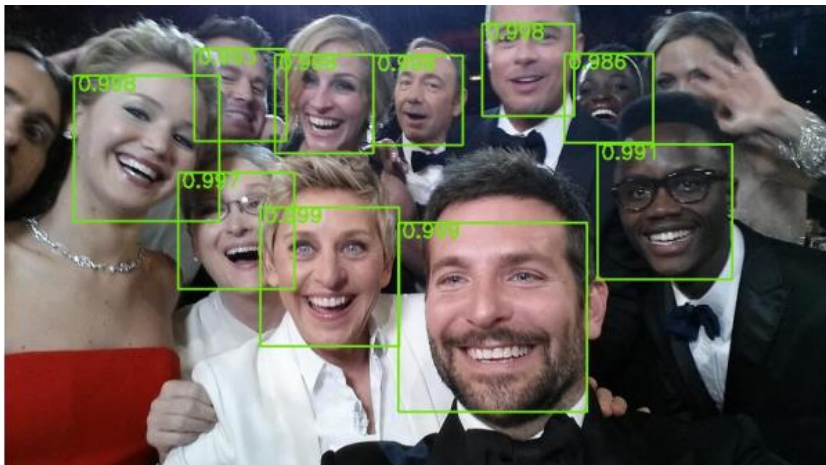
- Object Detection: Task Specification
- Viola-Jones Face Detector
- HOG + SVM
- DPM

Classification v.s. Detection



Detect X (X = Face, pedestrian, traffic sign, text etc.)

- We will omit 3-D detection

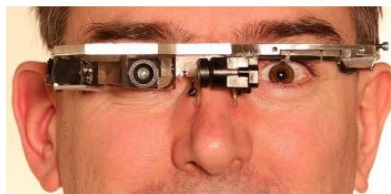


More Applications!

Robotics



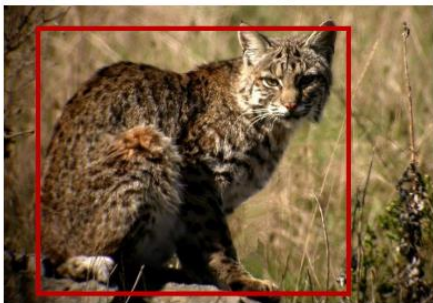
Augmented Reality



Mobile



Wildlife Monitoring



Self-Driving Cars



Computer Does Better! – in many scenarios

Funny Nikon ad: ""The Nikon S60 detects up to 12 faces."



Challenges

view point variation



deformation



Xu, Beihong 1943

and small things
from Apple.
(Actual size)

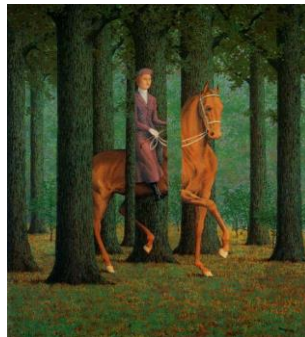
scale



illumination



occlusion



Magritte, 1957

background clutter



Klimt, 1913

Object Detection by Human Brain

- Find Waldo!





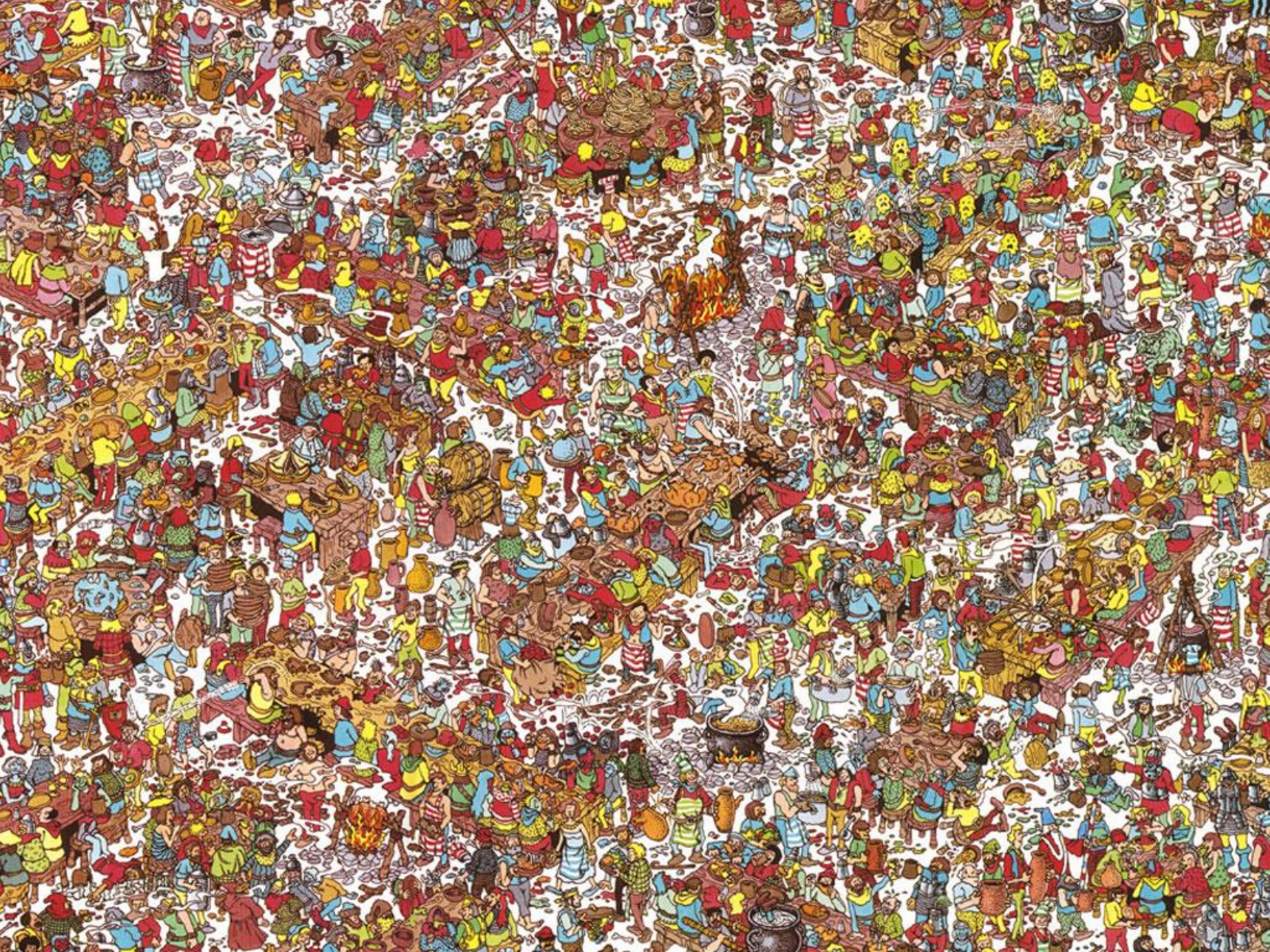


Image Classification v.s. Object Detection

- Detection-by-classification
- Sliding window detector must evaluate tens of thousands of location/scale combinations
- Faces are rare: 0–10 per image
 - For computational efficiency, we should try to spend as little time as possible on the non-face windows
 - A megapixel image has $\sim 10^6$ pixels and a comparable number of candidate face locations
 - To avoid having a false positive in every image, our false positive rate has to be less than 10^{-6}

Image Classification v.s. Object Detection

- Key challenges
 - Image classification: discriminative feature representation
 - Object detection: balancing accuracy and efficacy
- Object detection is more difficult than image classification

The Viola-Jones Face Detector

- A seminal approach to real-time object detection
- Training is slow, but detection is very fast
- Key ideas
 - *Integral images* for fast feature evaluation
 - *Boosting* for feature selection
 - *Attentional cascade* for fast rejection of non-face windows

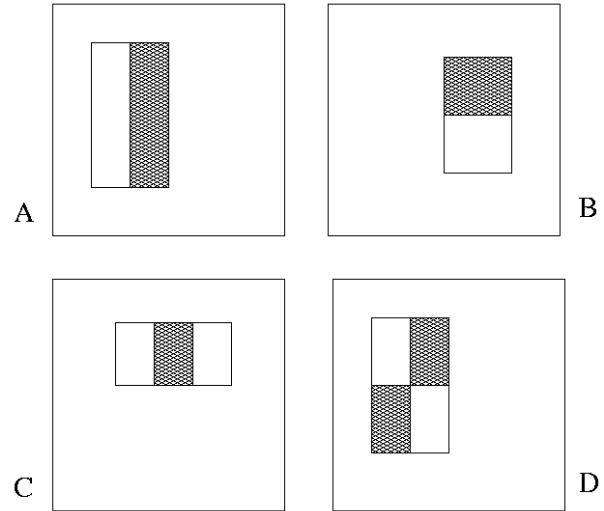
“...This face detection system is most clearly distinguished from previous approaches in its ability to detect faces extremely rapidly. Operating on 384 by 288 pixel images, faces are detected at 15 frames per second on a conventional 700 MHz Intel Pentium III...”

P. Viola and M. Jones. Rapid object detection using a boosted cascade of simple features. CVPR 2001.

P. Viola and M. Jones. Robust real-time face detection. IJCV 57(2), 2004.

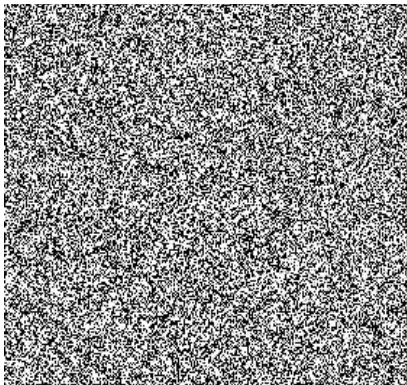
Image Features

“Rectangle filters”



$$Value = \sum (\text{pixels in white area}) - \sum (\text{pixels in black area})$$

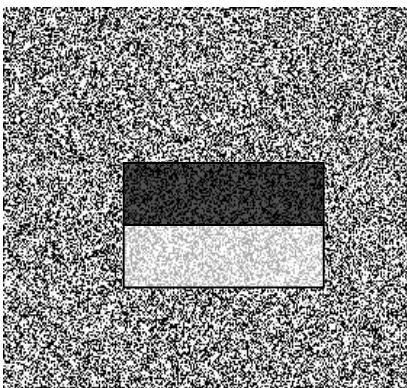
Example



Source

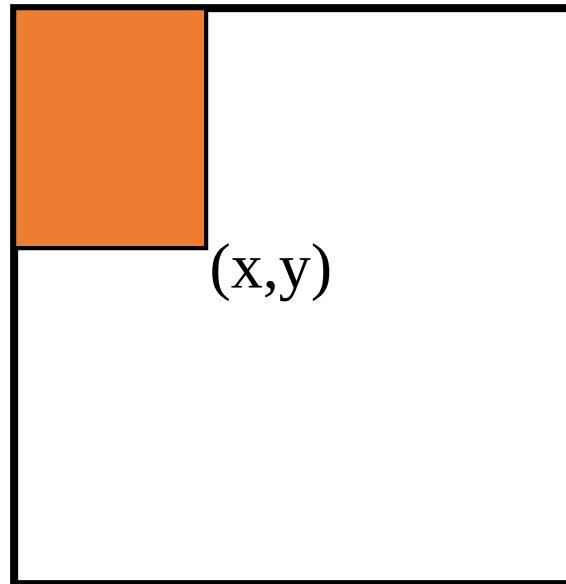


Result



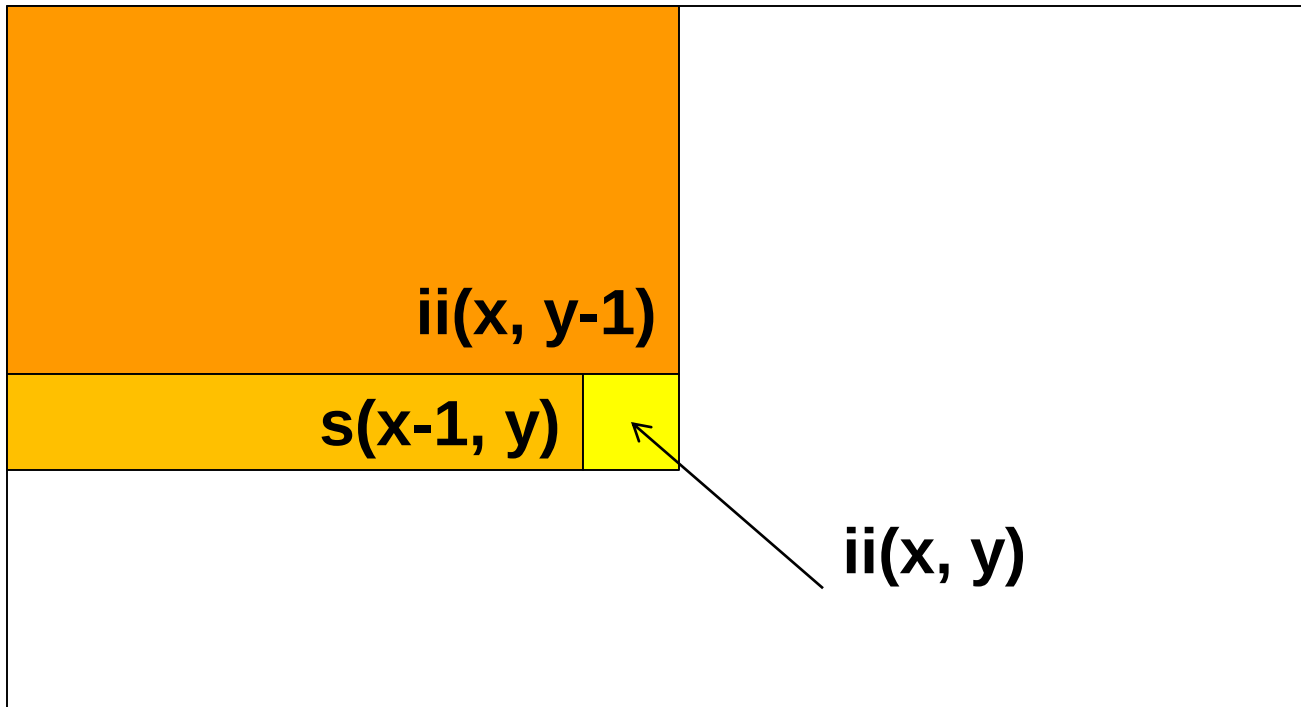
Fast computation with integral images

- The *integral image* computes a value at each pixel (x,y) that is the sum of the pixel values above and to the left of (x,y) , inclusive
- This can quickly be computed in one pass through the image



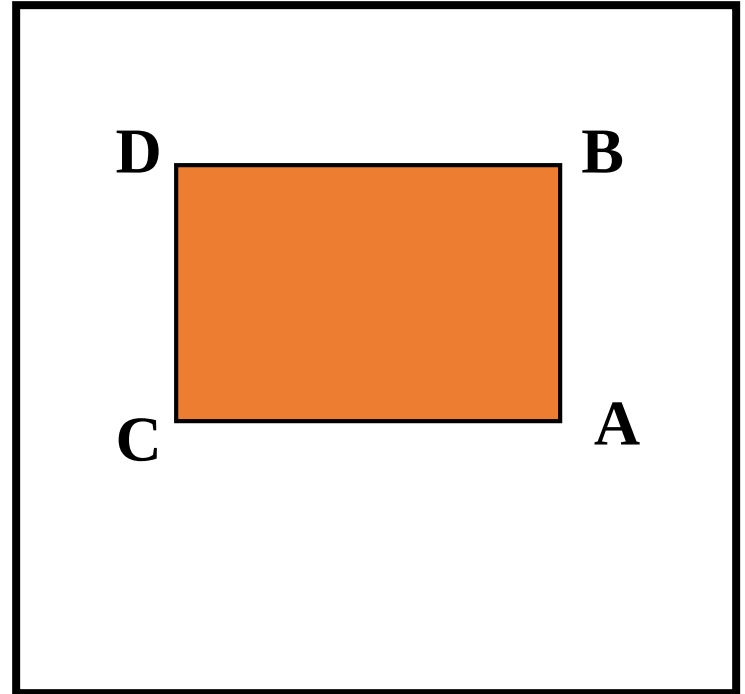
Computing the integral image

- Cumulative row sum: $s(x, y) = s(x-1, y) + i(x, y)$
- Integral image: $ii(x, y) = ii(x, y-1) + s(x, y)$
- MATLAB: `ii = cumsum(cumsum(double(i)), 2);`

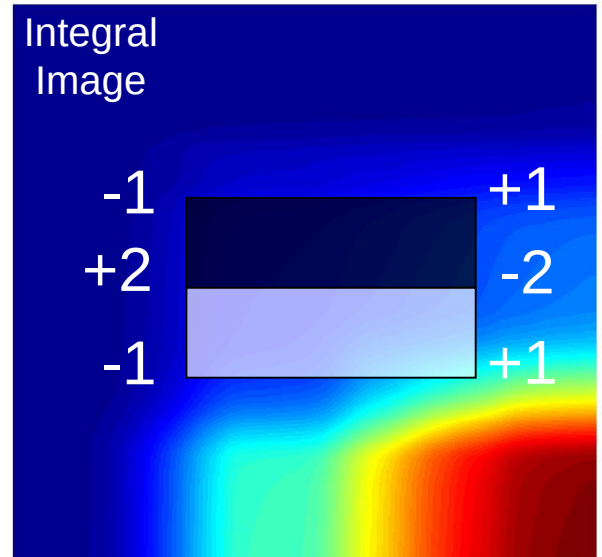
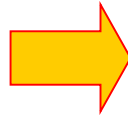


Computing sum within a rectangle

- Let A,B,C,D be the values of the integral image at the corners of a rectangle
- Then the sum of original image values within the rectangle can be computed as:
$$\text{sum} = A - B - C + D$$
- Only 3 additions are required for any size of rectangle!

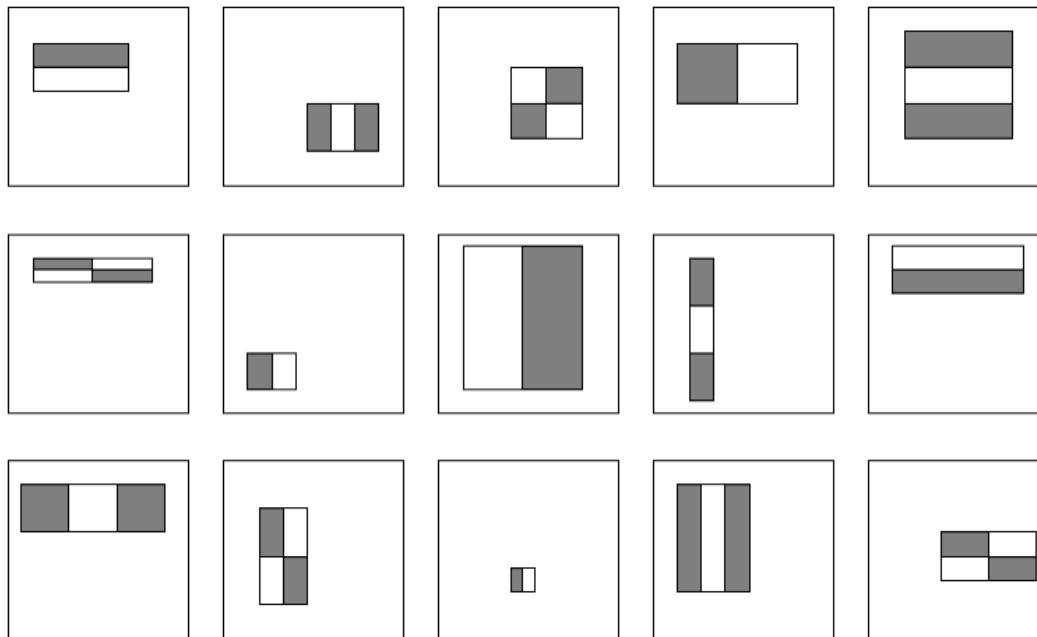


Example



Feature Selection

- For a 24x24 detection region, the number of possible rectangle features is ~160,000!
- Create a good classifier using just a small subset of all possible features!



Boosting

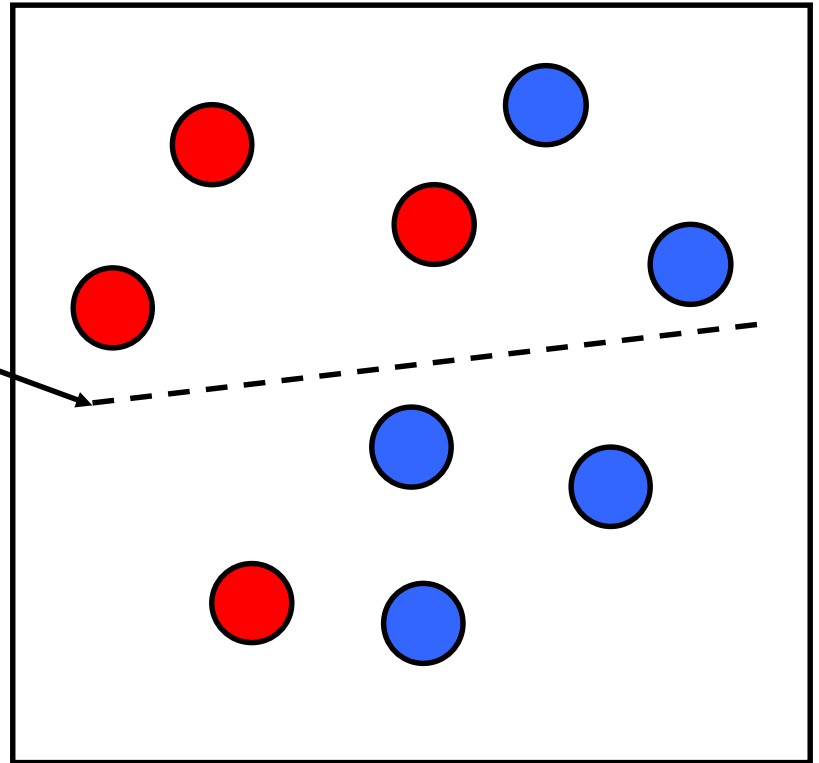
- Boosting is a classification scheme that works by combining *weak learners* into a more accurate ensemble classifier
 - A weak learner need only do better than chance
- Training consists of multiple *boosting rounds*
 - During each boosting round, we select a weak learner that does well on examples that were hard for the previous weak learners
 - “Hardness” is captured by weights attached to training examples

Y. Freund and R. Schapire, [A short introduction to boosting](#), *Journal of Japanese Society for Artificial Intelligence*, 14(5):771-780, September, 1999.

Boosting illustration

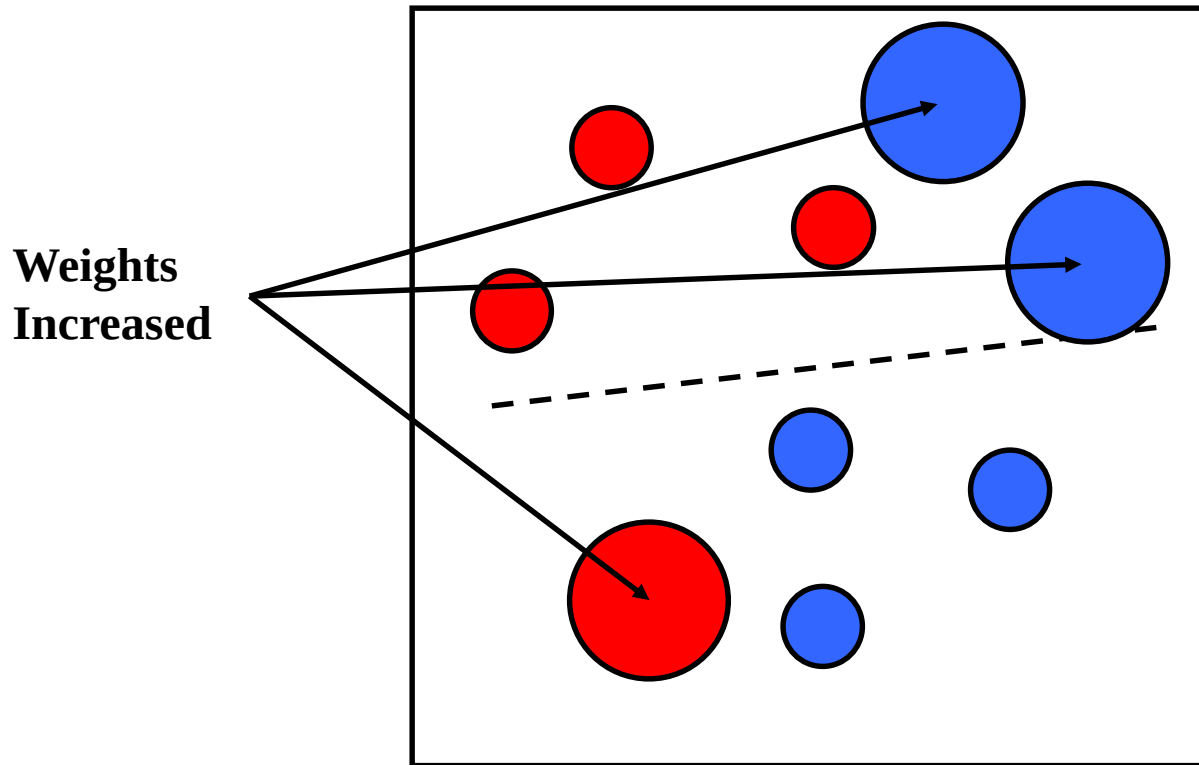
Initially, weight each training example equally

**Weak
Classifier 1**

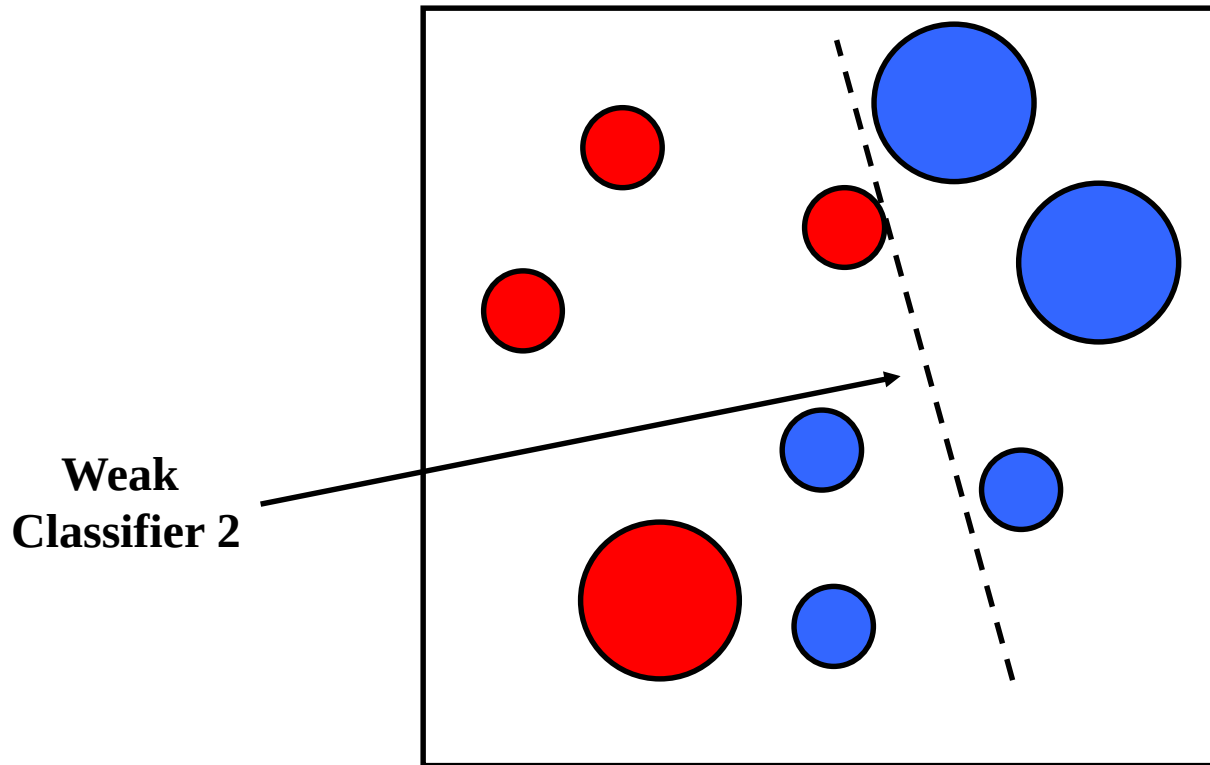


Boosting illustration

Raise the weights of training examples misclassified by current weak learner

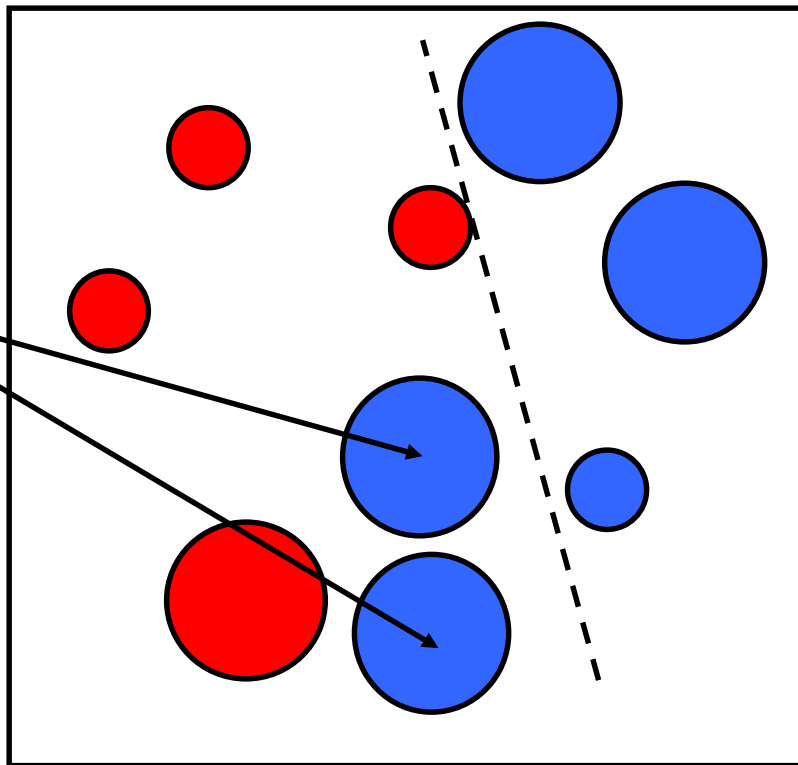


Boosting illustration

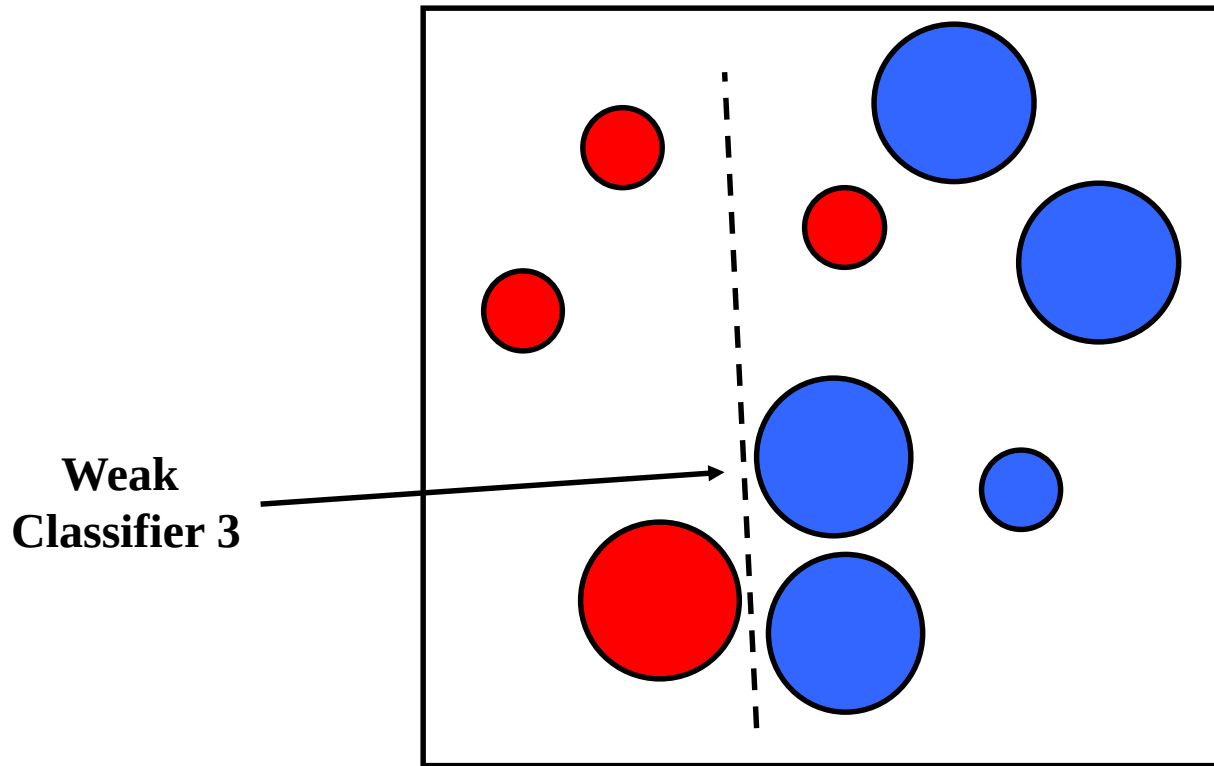


Boosting illustration

**Weights
Increased**

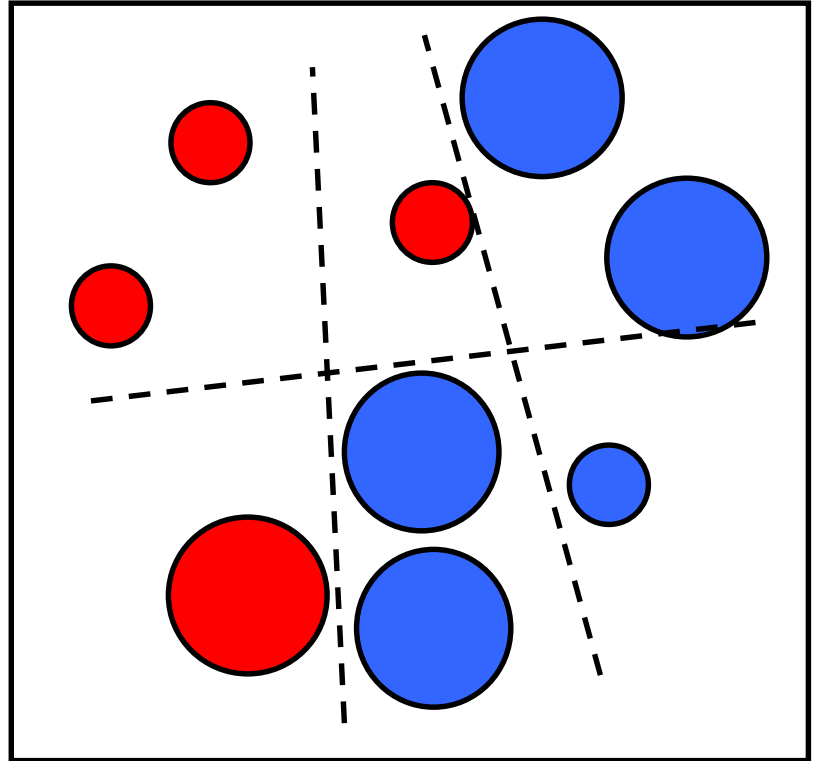


Boosting illustration



Boosting illustration

**Final classifier is
a combination of weak
classifiers**



Training procedure

- Initially, weight each training example equally
- In each boosting round:
 - Find the weak learner that achieves the lowest *weighted* training error
 - Raise the weights of training examples misclassified by current weak learner
- Compute final classifier as linear combination of all weak learners (weight of each learner is directly proportional to its accuracy)
- Exact formulas for re-weighting and combining weak learners depend on the particular boosting scheme (e.g., AdaBoost)

Y. Freund and R. Schapire, [A short introduction to boosting](#), *Journal of Japanese Society for Artificial Intelligence*, 14(5):771-780, September, 1999.

Boosting for face detection

- Define weak learners based on rectangle features

$$h_t(x) = \begin{cases} 1 & \text{if } p_t f_t(x) > p_t \theta_t \\ 0 & \text{otherwise} \end{cases}$$

Diagram illustrating the weak learner function $h_t(x)$ based on rectangle features:

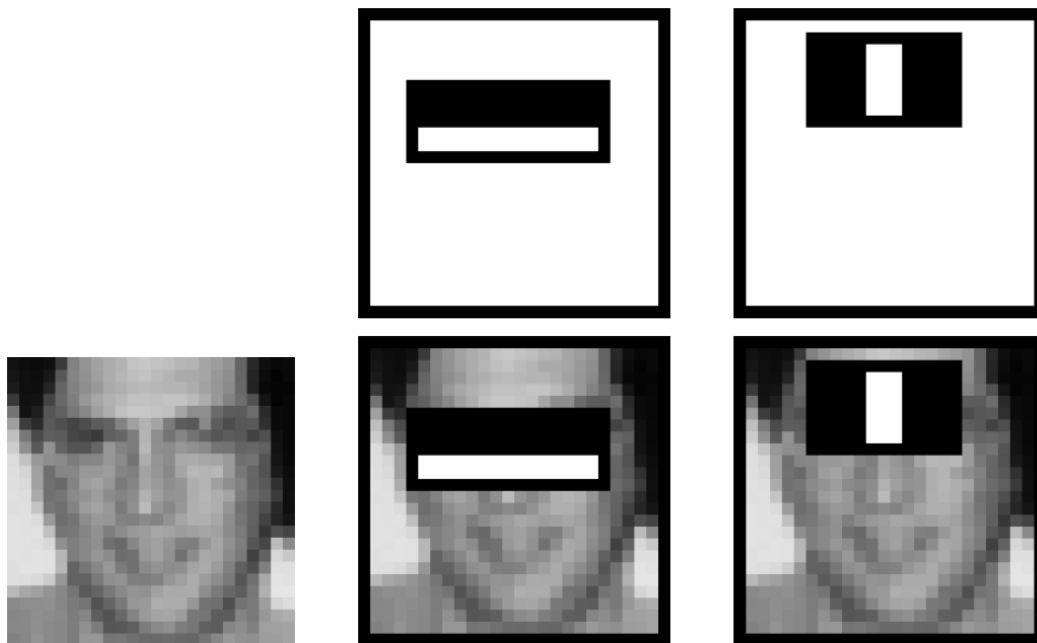
- $h_t(x)$: window
- $f_t(x)$: value of rectangle feature
- p_t : parity
- θ_t : threshold

Boosting for face detection

- Define weak learners based on rectangle features
- For each round of boosting:
 - Evaluate each rectangle filter on each example
 - Select best threshold for each filter
 - Select best filter/threshold combination
 - Reweight examples
- Computational complexity of learning: $O(MNK)$
 - M rounds, N examples, K features

Boosting for face detection

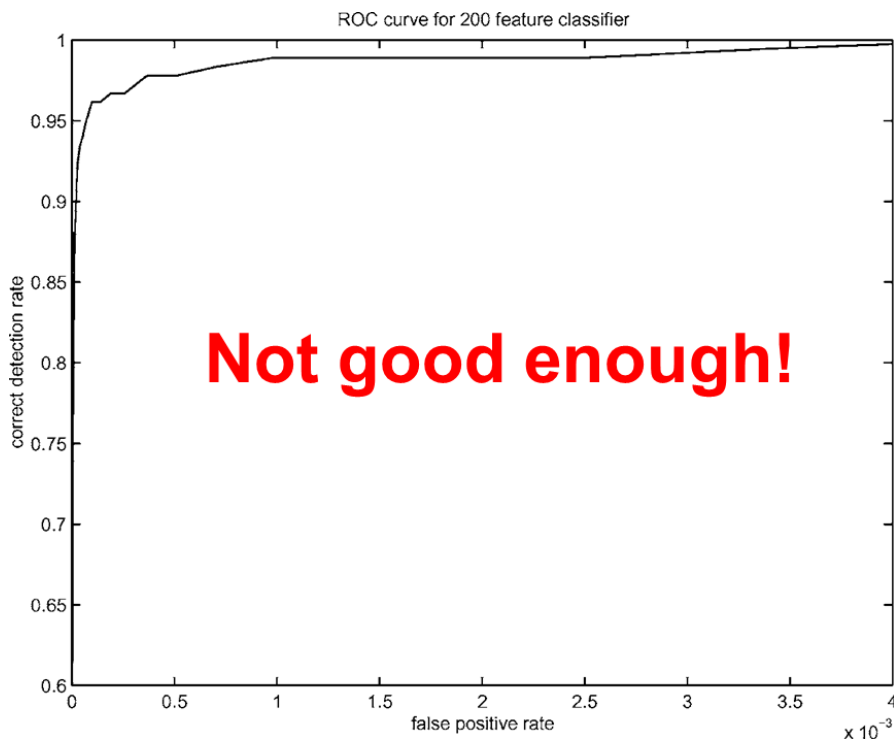
- First two features selected by boosting



This feature combination can yield 100% detection rate and 50% false positive rate

Boosting for face detection

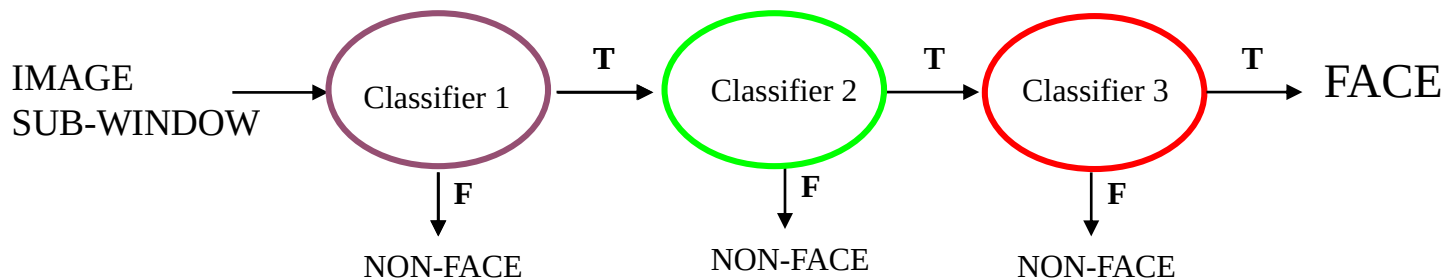
- A 200-feature classifier can yield 95% detection rate and a false positive rate of 1 in 14084



Receiver operating characteristic (ROC) curve

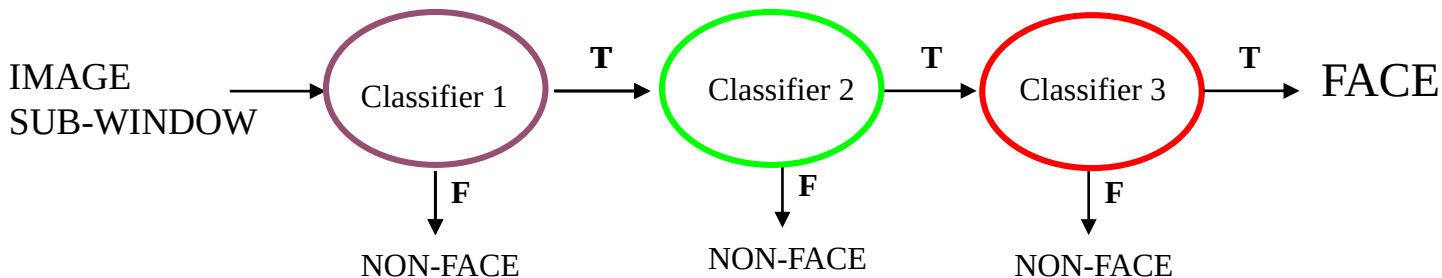
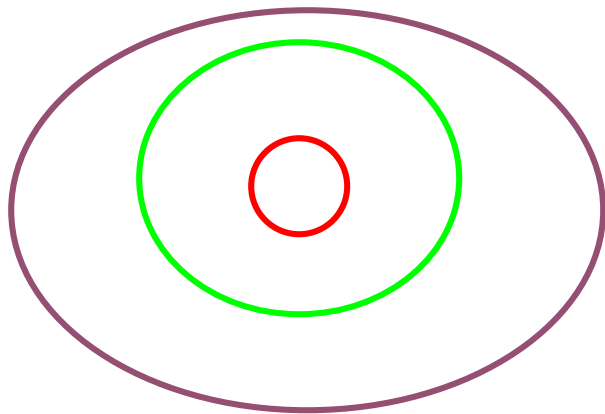
Attentional cascade

- We start with simple classifiers which reject many of the negative sub-windows while detecting almost all positive sub-windows
- Positive response from the first classifier triggers the evaluation of a second (more complex) classifier, and so on
- A negative outcome at any point leads to the immediate rejection of the sub-window

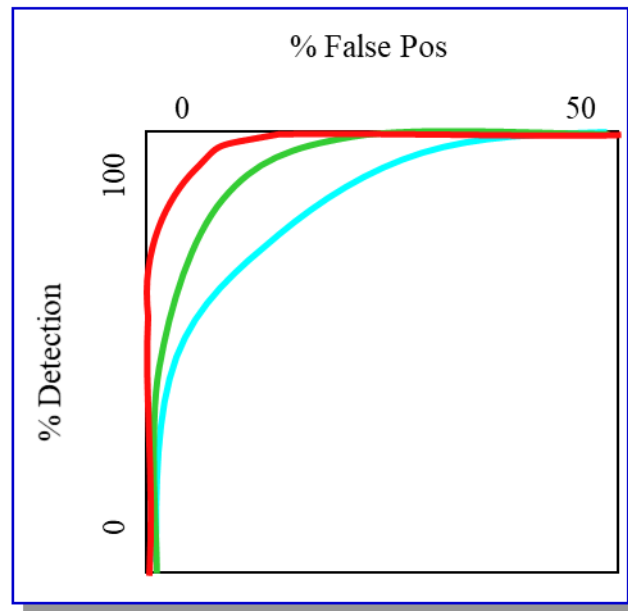


Attentional cascade

- Chain classifiers that are progressively more complex and have lower false positive rates:



Receiver operating characteristic



Attentional cascade

- The detection rate and the false positive rate of the cascade are found by multiplying the respective rates of the individual stages
- A detection rate of 0.9 and a false positive rate on the order of 10^{-6} can be achieved by a 10-stage cascade if each stage has a detection rate of 0.99 ($0.99^{10} \approx 0.9$) and a false positive rate of about 0.30 ($0.3^{10} \approx 6 \times 10^{-6}$)

Training the cascade

- Set target detection and false positive rates for each stage
- Keep adding features to the current stage until its target rates have been met
 - Need to lower AdaBoost threshold to maximize detection (as opposed to minimizing total classification error)
 - Test on a *validation set*
- If the overall false positive rate is not low enough, then add another stage
- Use false positives from current stage as the negative training examples for the next stage

The implemented system

- Training Data

- 5000 faces
 - All frontal, rescaled to 24x24 pixels
- 300 million non-faces
 - 9500 non-face images
- Faces are normalized
 - Scale, translation

- Many variations

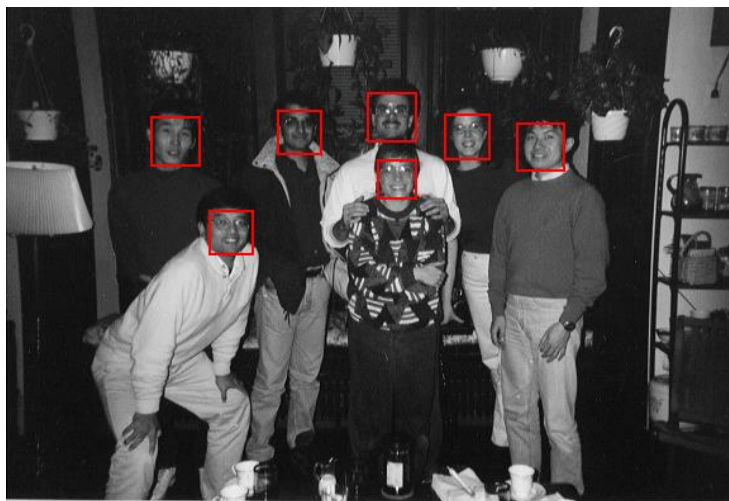
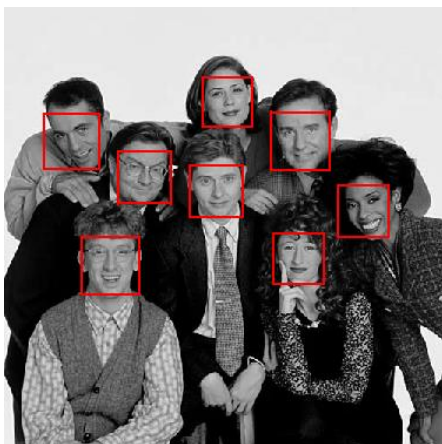
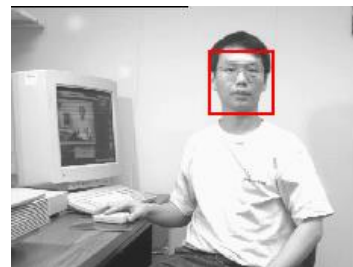
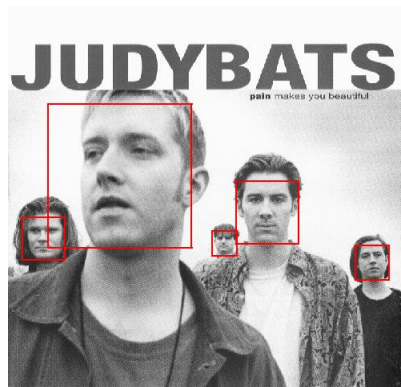
- Across individuals
- Illumination
- Pose



System performance

- Training time: “weeks” on 466 MHz Sun workstation
- 38 layers, total of 6061 features
- Average of 10 features evaluated per window on test set
- “On a 700 Mhz Pentium III processor, the face detector can process a 384 by 288 pixel image in about .067 seconds”
 - 15 Hz
 - 15 times faster than previous detector of comparable accuracy (Rowley et al., 1998)

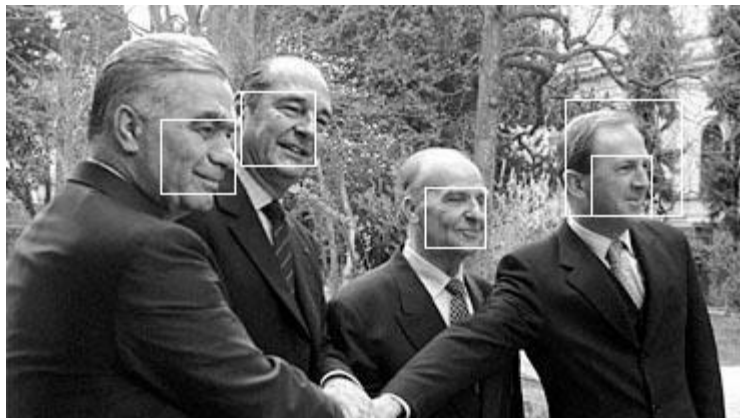
Output of Face Detector on Test Images



Other detection tasks

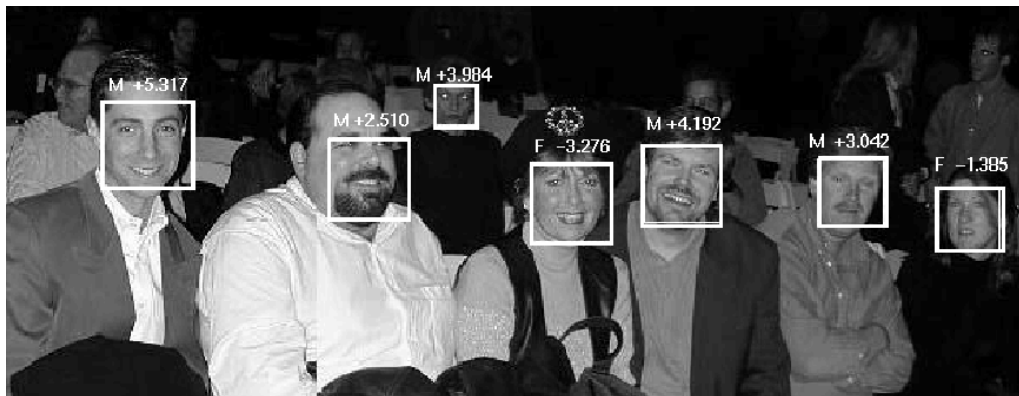


Facial Feature Localization

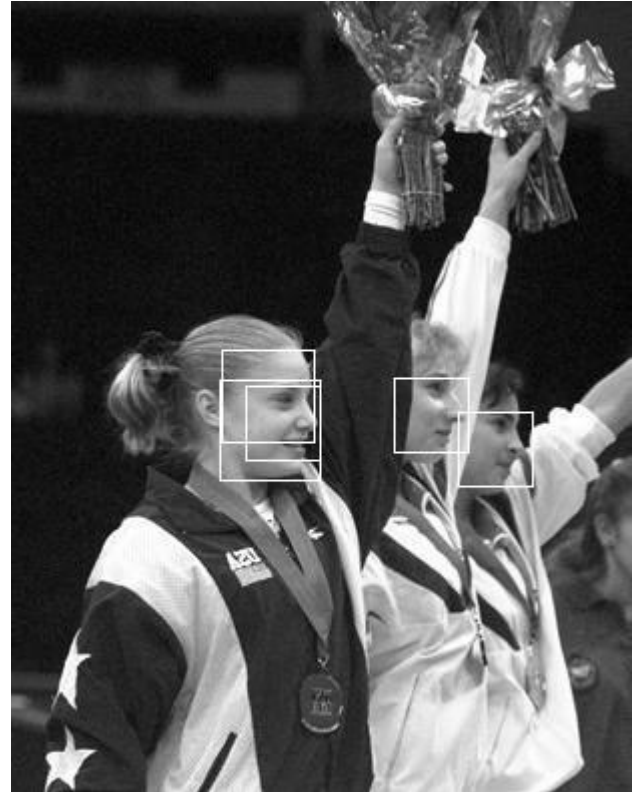


Profile Detection

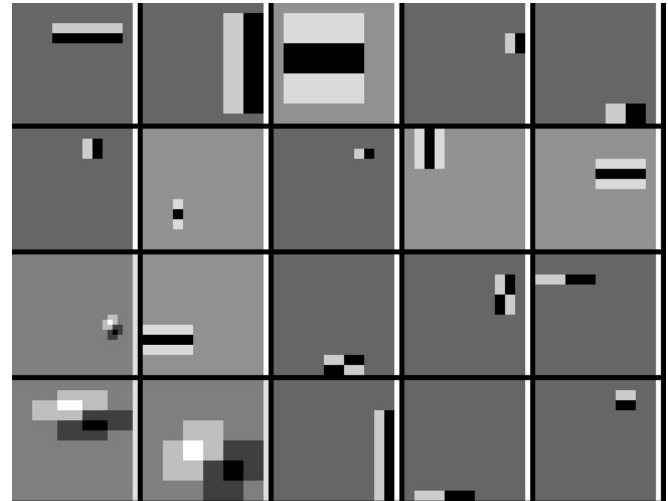
Male vs.
female



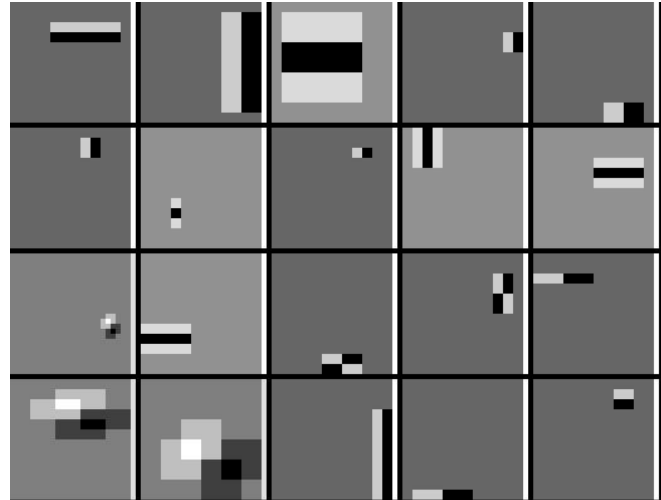
Profile Detection



Profile Features



Profile Features



Summary: Viola/Jones detector

- Rectangle features
- Integral images for fast computation
- Boosting for feature selection
- Attentional cascade for fast rejection of negative windows

HOG + SVM

- See Sanja Fidler's slides
- <http://www.cs.utoronto.ca/~fidler/CSC420.html>

DPM

- See Sanja Fidler's slides
- <http://www.cs.utoronto.ca/~fidler/CSC420.html>

Questions?

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